Record Room

EHR Media Intelligence Platform | Anass Bentaib | November 11, 2026

A Python/FastAPI pipeline cleans JSON and CSV exports, produces patient-specific FHIR R4 bundles, stores them in SQLite, and indexes documents and summaries with MiniLM and FAISS. The Tailwind interface provides ranked search, date/type filters, patient details and source links. The fixture contains 50 fictional documents for 10 patients.

TRADEOFFS AND DESIGN

Exact MRN matching and conflict quarantine favor avoiding incorrect patient merges over recall. Missing dates and demographics remain unknown. Summaries use LLM-selected, source-linked sentences instead of free paraphrase; this limits fluency but enables deterministic evidence checks. Cache keys include patient ID, bundle hash, provider, model and prompt version. Local embeddings and exact FAISS search keep the small dataset reproducible without an embedding API. Imports are offline snapshots; the server must restart after an import.

FHIR AND CLINICAL RESEARCH

Research covered DocumentReference versus DiagnosticReport, bundle reference resolution, administrative gender, unknown encounter class, and partial-date semantics. Date-only sources are not converted into invented instants. Current fhir.resources models target R4B, so the application additionally validates against the official R4 4.0.1 schema and checks references. Generated XHTML narratives preserve source text. HL7 validator 7.0.5 returned zero errors and zero warnings for all 10 bundles; 80 informational notices reflect disabled remote MIME-type terminology checks.

VALIDATION AND SUMMARY QUALITY

All 41 local tests passed, including actual MiniLM retrieval for three paraphrased queries. Tests cover cleaning, conflicts, FHIR errors, filters, cache invalidation, unsupported excerpts, negation fragments and word limits. The real OpenAI SDK is tested through an HTTP mock, including refusals and failures. No live API key was configured: displayed summaries are explicitly labeled deterministic demo excerpts. Live-model quality and clinician agreement remain unverified. The review rubric checks attribution, numbers, negation, chronology and coverage against linked sources.

Twenty-five warm POST /search requests over 50 documents measured 25.92 ms median, 35.04 ms p95 and 42.69 ms maximum on the development Mac, excluding model download/startup. Browser checks covered desktop/mobile search, filters, empty states, patient drawers and keyboard operation.

WITH MORE TIME

Add source-qualified identities, OCR, SMART-on-FHIR authentication, authorization, protected audit storage and atomic background imports. Extend validation with receiving-system profiles and a terminology server. Evaluate live summaries and retrieval with clinician-reviewed cases, including temporal contradictions, negation and prompt injection. Generic MiniLM similarity and exact excerpt matching do not establish clinical correctness.